

Searching Architecture and Precision for U-net based Image Restoration Tasks

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Abstract—Manually architecting Deep Neural Networks (DNNs) has led to the success of Deep Learning in many domains. However, recent DNNs designed using Neural Architecture Search (NAS) have exceeded manually designed architectures and have significantly reduced the human effort to develop complex networks. Current works use NAS to identify a cell architecture constrained by a fixed order of operations that is then replicated throughout the network. The constraints potentially limit the effectiveness of NAS in converging on a more efficient DNN architecture. In the first part of our paper, we propose “Operation Search,” a search on an enlarged topological space for U-net and its variants that retain efficiency. The idea is to allow for custom cells (operations and their sequence) at various levels of the network to maximize image quality while being sensitive to computation cost. In the second part of our paper, we propose custom quantization at various levels resulting in a mixed-precision network. Additionally, we increase the search efficiency by constraining the search space to use the same precision for both weights and activations at any level. This does not result in computational inefficiency because it matches the operand precisions supported by Tensor Core enabled GPUs.

Index Terms—Neural Architecture Search, U-net, Image Reconstruction, Mixed Precision

I. INTRODUCTION

The initial Convolutional Neural Networks (CNNs) [1], [2] were manually designed by domain experts, based on a deep understanding of image processing. The trial and error process of several architectures for a task is time-consuming and labor-intensive. Neural architecture search (NAS) has been very efficient in the last few years, whose goal is to build a Neural Network without significant human intervention. The primary objective of NAS [3], [4] is to construct a search space that contains all possible network configurations for a given backbone network and search for the efficient architecture with respect to various constraints like accuracy, size, and latency. The most successful one turned out to be DARTS [5], which starts by constructing and training a supernet of all possible combinations, followed by sampling a path within the supernet to be the final searched network.

The first and important step in the differentiable search process is to pick a backbone/skeleton on which we search for an architecture. We choose the U-Net [6], [7] model that is very well suited for a variety of applications which has a low compute cost. The U-net architecture consists of an Encoder

side (Down-Sampling Cells) and a Decoder side (Up-Sampling Cells), as shown in Fig. 1. The downsampling cell (left path in Fig. 1) halves the input feature map, whereas an upsampling cell (right path in Fig. 1) doubles the feature map. The decoder cells are fused with the skip connections from the encoder side, which improves the ability to segment details.

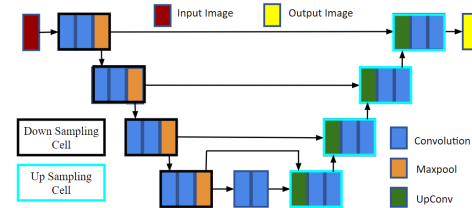


Fig. 1: U-net Model of Depth 4

We work on designing efficient search space, specific to the U-net model with experiments on Image Restoration tasks such as Super-Resolution and Denoising, with differentiable search as a search strategy [5]. Super-Resolution (SR) [8] is a task to design methods to recover High Resolution (HR) images from Low Resolution (LR) ones. Denoising [9] is a task to restore an HR or clean image from a distorted image.

Quantization is a promising solution to solve the difficulties faced by modern CNNs in real-time deployment. The main objective of quantization is to convert the high-precision (floating-point 32) weights to low-precision (Integer 8/4/1). Uniform Quantization [10] is the process of quantizing all the layers in the network to the same precision. Int8/Int4 Quantization of CNN followed by fine-tuning proved to be very successful without significant loss in accuracy [10], [11].

Mixed Precision Quantization [12] strikes a balance between accuracy and low-precision implementation. This enables the optimization of bit-widths at the filter level and allows each layer to execute with different precision. The problem lies in deciding the precision for each layer in the network such that the cost of the model is reduced through quantization while maintaining the accuracy. For a network consisting of “L” layers, and “N” bit-widths, we can have N^L different network choices. It is very difficult/impossible to try out every possible combination and find the optimal bit-width of every layer. Hence, we need an automatic bit allocation method for